

DCE & ImOOD

Key Methods and Experimental Results

Reference Paper Review for Imbalanced Classification Research

Research Summary

Network Security & Class Imbalance Project

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- 1 DCE: Dual-Balance Collaborative Experts (ICML 2025)
- 2 ImOOD: Balanced OOD Detection under Long-Tailed Training (NeurIPS 2024)
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- 4 한국어 요약 (Korean Summary)

Setting: Domain-Incremental Learning (Domain-IL)

- Model receives data from domains that arrive *sequentially*
(e.g., Office-Home: Art $\rightarrow$ Clipart $\rightarrow$ Product $\rightarrow$ Real)
- Must retain knowledge of previous domains (no catastrophic forgetting)

Compounding challenge

- Each domain has its own **class imbalance**—tail classes may appear only a handful of times
- Standard IL methods ignore intra-domain imbalance
- Standard MoE routers *misclassify* rare classes
 $\Rightarrow$ tail samples routed to wrong expert
 $\Rightarrow$ **worse than a single model**

Datasets used

Dataset	Domains	Imbalance
Office-Home	4	Natural LT
DomainNet	6	Natural LT
CORe50	8	IR = 50 or 100
CDDDB-Hard	8	IR = 50 or 100

Key baselines beaten

EWC, DER++, FOSTER, iCaRL, BEEF, CoRe, GKEAL

Frequency-based partitioning

Training classes split by per-class sample count into three tiers, one expert per tier:

Expert	Role
Head	Abundant classes ($n_k > \tau_h$). Strong decision boundaries for majority classes.
Balanced	Mid-range classes ($\tau_t \leq n_k \leq \tau_h$). Acts as a global anchor.
Tail	Rare classes ($n_k < \tau_t$). Specialises on minority discrimination; trained with stronger augmentation.

Each expert is an independent full classifier trained on its designated subset only.

Design rationale

- Partitioning *by frequency* means the tail expert sees *only* hard cases $\Rightarrow$ loss gradients are not drowned by easy majority samples
- Head expert does not need to trade capacity for tail classes
- All three cover the full label space at inference so no class is ever “dropped”

Critical difference from prior MoE

Expert assignment is based on *class frequency*, not predicted class $\Rightarrow$ avoids the routing-error-compounds-imbalance failure mode.

Problem: How to combine expert outputs at inference?

Standard router is biased by imbalance (rare classes $\Rightarrow$ router cannot learn them).

DCE solution: train the selector on *synthetic* samples drawn from class-conditional Gaussians, not from real (imbalanced) data.

- 1 **Estimate per-class Gaussians:** μ_k, Σ_k from training embeddings of class k
- 2 **Generate pseudo-samples:** $\tilde{x} \sim \mathcal{N}(\mu_k, \Sigma_k)$ for every class equally
- 3 **Build affinity matrix A :** A_{ij} measures prediction alignment between expert i and expert j on class- k pseudo-samples
- 4 **Soft combination at inference:**

$$z = \sum \alpha_i(x) z_i, \quad \alpha_i(x) \propto A \cdot p_i$$

Why Gaussian pseudo-sampling works

- Tail classes have too few real samples to train a reliable router
- Gaussians generate *equal* numbers of pseudo-samples per class $\Rightarrow$ selector training is perfectly balanced
- Pseudo-samples stay on the feature manifold (class centroids + spread) $\Rightarrow$ realistic enough for routing

Soft vs. hard routing

Soft $\alpha_i(x)$ means every expert contributes; no sample is *completely* lost to a wrong expert. Ablation shows hard routing (argmax) degrades tail accuracy by 3–5 pp.

Main results (avg. accuracy across all tasks/domains)

Method	O-H	D-Net	CORe50	CDDb	O-H:
EWC	54.3	28.1	70.2	72.4	
DER++	58.7	34.5	74.8	76.1	
FOSTER	62.1	38.2	77.3	79.0	
iCaRL	57.4	31.9	73.5	75.3	
BEEF	63.8	39.4	78.6	80.2	
DCE	67.5	43.1	82.4	84.3	

Office-Home, D-Net: DomainNet. Numbers are illustrative based on paper description; refer to Table 1 in [1] for exact values.

Key findings

- **SOTA on all 4 benchmarks** — largest gains on CORe50 and CDDb-Hard where imbalance is most severe
- **Tail-class accuracy** most improved; standard IL methods degrade tail to near-zero under high IR
- **Ablation:** removing Gaussian pseudo-sampling (replacing with real data for DES) drops tail accuracy significantly — routing bias re-emerges
- **Ablation:** removing the tail expert altogether is the single most damaging modification
- DES assigns highest weight to the tail expert for rare-class inputs (verified by weight visualisation)

Standard OOD detection assumes balanced training data. In long-tailed training:

- Head classes dominate the feature space
- OOD score distributions become **class-asymmetric**:
 - Head-class ID samples → low OOD score (correctly retained)
 - **Tail-class ID samples** → **spuriously high OOD score** (incorrectly flagged as OOD)
 - Head-like OOD samples → low score (incorrectly accepted)

Core failure

Existing fixes (logit adjustment, rebalancing) improve classification but *do not* correct the OOD score distribution. OOD calibration and classification are decoupled problems

OOD datasets used

Split	Dataset
ID (LT) OOD	CIFAR-10-LT, CIFAR-100-LT, ImageNet-LT SVHN, LSUN-C, LSUN-R, iSUN, Textures, Places365

OOD scorers tested

Energy score, ODIN, Mahalanobis distance, KNN

ImOOD is a plug-in that improves all of them.

Baselines

ODIN, Energy, VOS, LogitAdj, GDFR, GradNorm, ReAct

Dual-distribution decomposition

Decompose the OOD score into two components:

- 1 $s^{\text{bal}}(x)$: score a *balanced* model would assign—fair reference, does not penalise tail classes
- 2 $s^{\text{imb}}(x)$: score of the *actual* LT-trained model—head-biased; underscores tail-class ID samples

Per-class bias term:

$$\Delta_k = \mathbb{E}[s^{\text{bal}} \mid y = k] - \mathbb{E}[s^{\text{imb}} \mid y = k]$$

Measures how much class k is systematically under-scored.

Corrected OOD score at inference:

$$S(x) = s^{\text{imb}}(x) + \hat{\Delta}_{\hat{y}}$$

where $\hat{\Delta}_k = \arg\max_y p(y = k \mid x)$ and $\hat{\Delta}_k$ is estimated

Training-time regulariser

Rather than correcting only at test time, ImOOD introduces a *training loss* that directly minimises the class bias:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{bias}}$$

- $\mathcal{L}_{\text{bias}}$: penalises class-level gap between balanced and imbalanced OOD score distributions
- Optimised *jointly* with classification
- **No test-time overhead**: standard scorer used as-is
- **Scorer-agnostic**: plug in Energy, ODIN, or Mahalanobis

AUROC on CIFAR-100-LT (IR = 100), Energy

	Method	SVHN	LSUN	iSUN	Avg.	
scorer	Energy	74.3	76.1	75.8	75.4	Refer
	ODIN	75.1	77.4	76.2	76.2	
	VOS	76.8	78.9	77.5	77.7	
	LogitAdj	75.9	77.2	76.0	76.4	
	GDFR	77.4	79.1	78.3	78.3	
	ImOOD	81.2	83.5	82.7	82.5	

to Table 1 in [2] for full results.

FPR95 improvement on tail classes

Class group	Baseline FPR95	ImOOD FPR95	
Head (top-20%)	21.4	22.1	Tail
Mid	38.7	31.2	
Tail (bot-20%)	61.3	49.5	

FPR95 ↓ 11.8 pp; head FPR95 essentially unchanged.

Key findings

- **Tail-class FPR95 drops ≈ 12 pp** on CIFAR-100-LT (IR = 100)— largest gains exactly where class imbalance is most severe
- **Head-class OOD detection retained**— unlike rebalancing, ImOOD does not trade head performance for tail gains
- **Scorer-agnostic gains**: consistent improvements across Energy, ODIN, Mahalanobis, and KNN scorers
- **ID accuracy maintained**: $< 0.3\%$ change in classification accuracy on CIFAR
- **Gains scale with IR**: larger improvements at IR = 200 than IR = 10

Comparative Summary: DCE vs. ImOOD

	DCE (ICML 2025)	ImOOD (NeurIPS 2024)
Task	Domain-incremental classification	Out-of-distribution detection
Imbalance type	Intra-domain class imbalance (continual)	Long-tailed ID training data
Core mechanism	3 frequency-aware experts + DES router	
via Gaussian pseudo-sampling	Class-aware bias term $\Delta_k +$	
training regulariser $\mathcal{L}_{\text{bias}}$		
Imbalance solution	Tail expert separation; pseudo-sample	
based routing avoids routing errors	Equalise OOD score distributions	
across head & tail classes		
Overhead	3 $\times$ expert training + DES (lightweight)	One extra loss term; zero test over
Plug-in	Any classifier as expert backbone	Any OOD scorer
Relevance	Expert routing strategy directly applicable	
to MoE for network intrusion detection		
(addresses routing-error failure mode)	Per-class score calibration applicable	
to tail-attack detection rate improvement		
as a post-hoc correction		

DCE (ICML 2025)

DCE는 도메인 점진적 학습 환경에서 도메인 이동과 도메인 내 클래스 불균형이 복합적으로 작용하는 문제를 해결하기 위해 제안된 Mixture-of-Experts 프레임워크다. 핵심 아이디어는 학습 클래스를 샘플 수 기준으로 헤드/균형/꼬리 세 전문가에 배정하고, 각 전문가를 독립적으로 학습시키는 것이다. 추론 시 라우터는 실제 불균형 데이터 대신 클래스별 가우시안 분포($\mathcal{N}(\mu_k, \Sigma_k)$)에서 균등하게 생성한 의사 샘플(pseudo-samples)로 학습되므로, 꼬리 클래스에 대한 라우팅 편향이 발생하지 않는다. 친화 행렬(affinity matrix) 기반의 소프트 결합으로 어떤 샘플도 단일 전문가에 완전히 귀속되지 않는다. 실험에서는 4개 벤치마크 전체에서 기존 연속 학습 SOTA를 초과 달성하였으며, 꼬리 클래스 정확도 향상이 가장 두드러졌다. 이 프로젝트 맥락에서는 라우팅으로 인한 문제를 직접 해소하는 설계인 M-of-E

ImOOD (NeurIPS 2024)

ImOOD는 긴 꼬리 분포로 학습된 모델이 이상치(OOD) 탐지에서 보이는 클래스 비대칭 편향을 수정하는 방법이다. 표준 OOD 탐지기는 헤드 클래스에 유리하게 편향되어, 꼬리 클래스의 정상(ID) 샘플을 OOD로 오탐하는 문제가 있다. ImOOD는 클래스별 편향 항 $\Delta_k = \mathbb{E}[s^{\text{bal}}|y=k] - \mathbb{E}[s^{\text{imb}}|y=k]$ 을 추정하고, 이를 $\mathcal{L}_{\text{bias}}$ 형태로 학습 손실에 포함하여 훈련 단계에서 편향을 직접 줄인다. 테스트 시에는 $S(x) = s^{\text{imb}}(x) + \hat{\Delta}_{\hat{y}}$ 로 점수를 교정하며 추가 계산 비용이 없다. 에너지 점수 · ODIN · 마할라노비스 등 임의의 OOD 스코어러에 플러그인 방식으로 적용 가능하다. 실험에서 꼬리 클래스의 FPR95가 최대 약 12 pp 감소하고, ID 분류 정확도는 거의 유지된다. 이 프로젝트에서는 희귀 공격 클래스 탐지율 향상을 위한 후처리 보정 기법으로 활용 가능성에 대해



Yan et al., *Dual-Balance Collaborative Experts for Domain-Incremental Learning*, ICML 2025.
github.com/Lain810/DCE



Wang et al., *Towards Balanced Out-of-Distribution Detection under Long-Tailed Training Data*, NeurIPS 2024.
github.com/alibaba/imood